National
Qualifications
2018

X707/76/02

Biology
Section 1 — Questions

TUESDAY, 15 MAY
9:00 AM – 11:30 AM

Instructions for the completion of Section 1 are given on *page 02* of your question and answer booklet X707/76/01.

Record your answers on the answer grid on *page 03* of your question and answer booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

©

X7077602

SECTION 1 — 20 marks

Attempt ALL questions

1. The following substances are products of fermentation.

1 ATP

2 Lactate

3 Carbon dioxide

Which of these are products of fermentation in human muscle cells?

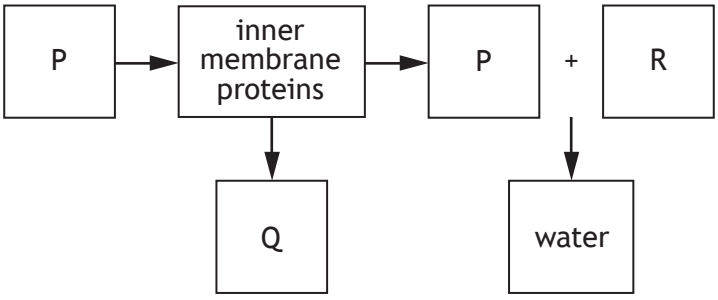
A 2 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

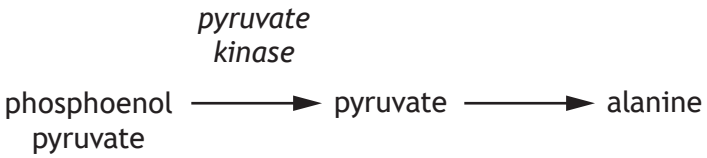
2. The diagram represents a stage of cellular respiration that occurs in a mitochondrion.



Which row in the table identifies substances P, Q and R?

	Substances		
	P	Q	R
A	ATP	hydrogen ions and electrons	oxygen
B	hydrogen ions and electrons	oxygen	ATP
C	oxygen	ATP	hydrogen ions and electrons
D	hydrogen ions and electrons	ATP	oxygen

3. Part of a metabolic pathway used by cells to produce the amino acid alanine is shown.



Alanine is a non-competitive, feedback inhibitor of the enzyme pyruvate kinase.

The following statements refer to the metabolic pathway.

1 Pyruvate kinase reduces the activation energy needed to convert phosphoenol pyruvate into pyruvate.

2 Phosphoenol pyruvate is the substrate for pyruvate kinase.

3 Alanine can bind to the active site of pyruvate kinase.

Which of these statements are correct?

A 1 and 2 only

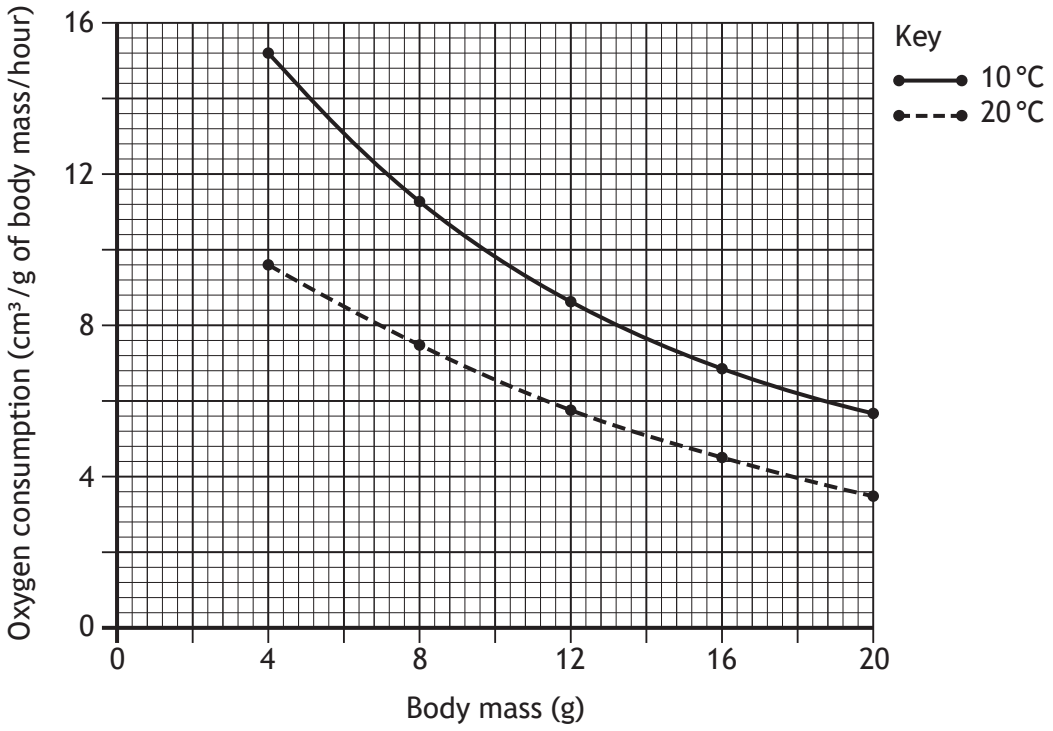
B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

[Turn over

4. Shrews are small mammals. The graph shows the relationship between body mass and oxygen consumption of shrews at two environmental temperatures.



Which of the following statements about this graph is correct?

A Shrews of greater mass consumed less oxygen.

B The optimum temperature for oxygen consumption was 10 °C.

C As environmental temperature increased oxygen consumption decreased.

D At 10 °C a 16 g shrew consumed 6·2 cm^3 of oxygen/g of body mass/hour.

5. Yeast cells contain the enzyme catalase which breaks down hydrogen peroxide to produce oxygen. An experiment was carried out into the effect of lead nitrate concentration on the activity of catalase.

Six flasks were set up. Each contained 25 cm^3 of hydrogen peroxide and 10 cm^3 of yeast suspension. 10 cm^3 of a different concentration of lead nitrate was then added to each flask. The volume of oxygen produced after 15 minutes was measured.

Identify the independent variable in this experiment.

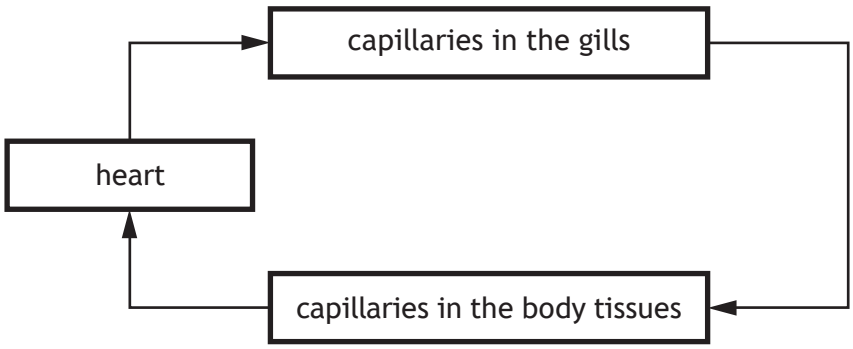
A Volume of lead nitrate

B Volume of oxygen produced

C Activity of catalase

D Concentration of lead nitrate

6. The diagram illustrates the circulatory system of a fish. The arrows indicate the direction of blood flow.



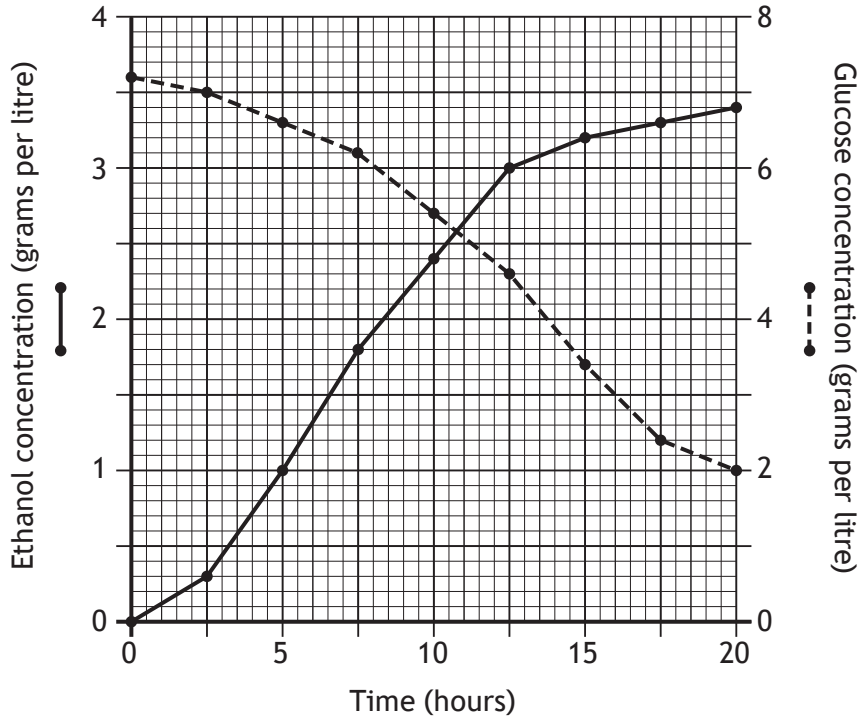
Which row in the table describes the type of circulatory system of a fish and the blood pressure in the capillaries in the gills and body tissues?

	Type of circulatory system	*Blood pressure in the capillaries in the gills*	*Blood pressure in the capillaries in the body tissues*
A	single	lower	higher
B	double	higher	lower
C	single	higher	lower
D	double	lower	higher

[Turn over

7. In an investigation into fermentation, yeast was grown in a flask of glucose solution for 20 hours at 20 °C.

The graph shows the concentrations of ethanol and glucose in the flask over the period of the investigation.



What was the glucose concentration when the ethanol concentration was 3·3 grams per litre?

A 1·2 grams per litre

B 2·2 grams per litre

C 2·4 grams per litre

D 6·6 grams per litre

8. The following statements describe stages in the Calvin Cycle (carbon fixation).

1 Carbon dioxide attaches to ribulose bisphosphate (RuBP) producing 3-phosphoglycerate (3PG).

2 3-phosphoglycerate (3PG) forms glyceraldehyde-3-phosphate (G3P).

3 Glyceraldehyde-3-phosphate (G3P) regenerates ribulose bisphosphate (RuBP).

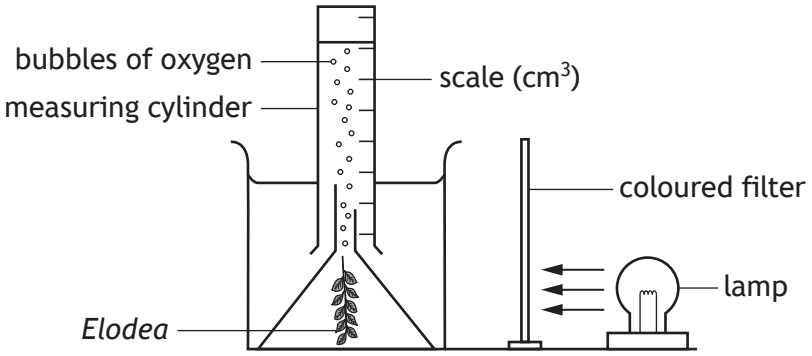
Which row in the table identifies the stage which is catalysed by RuBisCO and the stage which requires hydrogen?

	Catalysed by RuBisCO	*Requires hydrogen*
A	1	2
B	1	3
C	3	1
D	3	2

[Turn over

9. The diagram shows apparatus used in an investigation to measure the rate of photosynthesis in *Elodea* (pondweed) at different wavelengths of light.

Coloured filters were used to change the wavelength of the light. The volume of oxygen collected after 30 minutes was used to measure the rate of photosynthesis.



Suggested improvements to the investigation are shown.

1 Use a measuring cylinder with a narrower diameter.

2 Repeat the experiment several times and take averages.

3 Use a scale with more divisions.

Which of these suggestions would improve the accuracy of the results?

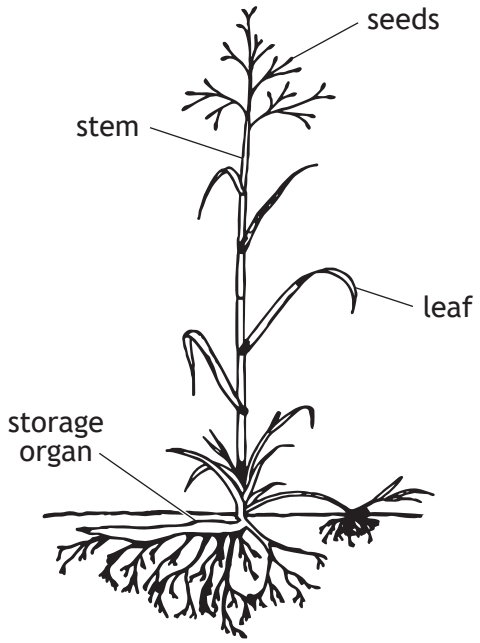
A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

10. The diagram shows a perennial weed found in agricultural land in Scotland.



Which feature of this weed indicates that it should be controlled by a systemic herbicide?

A Seeds

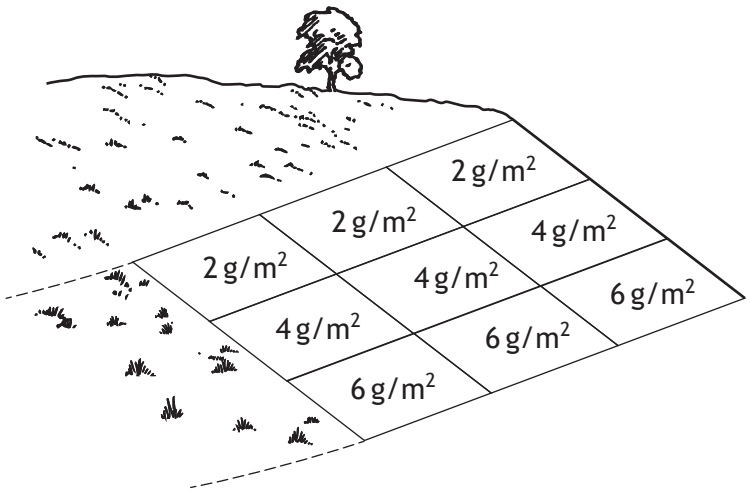
B Stem

C Storage organ

D Leaf

[Turn over

11. A field trial was carried out to investigate the effect of mass of phosphate fertiliser applied on the growth of barley. The barley was planted in plots of equal area on a hillside and fertiliser applied as shown in the diagram.



Which of the following procedures would improve the field trial design to take into account higher soil moisture levels at the bottom of the hill?

A Increase the range of phosphate fertiliser masses applied.

B Randomise the treatment plots.

C Increase the number of plots.

D Select another hillside.

12. Which of the following is an example of kin selection?

A Worker leafcutter ants raising young ants in their colony.

B A vampire bat regurgitating blood to feed an unrelated bat.

C A dominant lion feeding on a zebra kill before its offspring.

D A young orangutan spending a long period in parental care to learn complex social behaviours.

13. An experiment was carried out to investigate the growth rate of pigs. They were put into five groups of eight pigs, each with the same average initial body mass. Each group was fed a diet which contained either 0, 10%, 20%, 30% or 40% faba beans. The pigs were re-weighed each day for 40 days.

Which aspect of the experimental design increased reliability of the results?

A Five groups of pigs were used.

B The pigs were re-weighed each day for 40 days.

C Each group had the same average initial body mass.

D Each group contained eight pigs.

14. An investigation was carried out into the social hierarchy in a group of five hens, V, W, X, Y and Z. Hens establish dominance by pecking each other aggressively. The number of pecks given and received was recorded.

The results are shown in the table.

		Number of pecks given by each hen				
		V	W	X	Y	Z
Number of pecks received by each hen	V	-	-	-	10	-
	W	2	-	-	13	-
	X	6	8	-	7	-
	Y	-	-	-	-	-
	Z	11	10	5	4	-

The order of hierarchy from most dominant to least dominant hen is

A Z, V, X, W, Y

B Y, V, W, X, Z

C Z, X, W, V, Y

D Y, W, V, X, Z.

15. Each type of human cell has a different structure and function because

A only some of their genes are expressed

B they contain different genes

C some genes are lost during differentiation

D some genes are gained during differentiation.

[Turn over

16. The list describes some uses of stem cells.

1 Studying how cells differentiate
2 Researching the development of Parkinson's disease
3 Producing skin for skin grafts
4 Bone marrow transplants

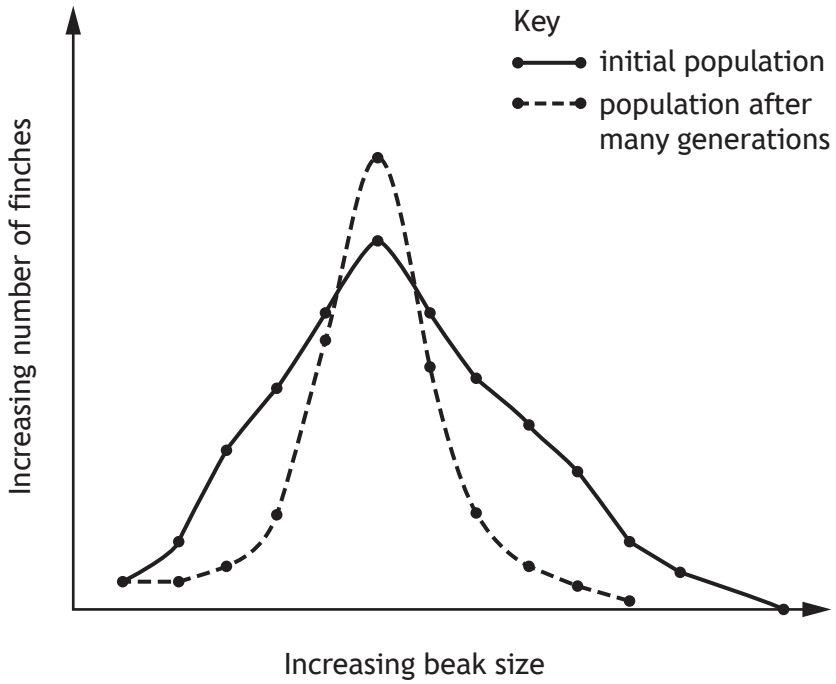
Which of these uses are **not** therapeutic?

A 1 only
B 1 and 2 only
C 2 and 4 only
D 3 and 4 only

17. Which of the following is an example of sexual selection?

A Koalas with resistance to disease surviving to reproduce.
B Peppered moths with the most effective camouflage avoiding predation.
C Plant breeders selecting barley cultivars to cross to improve grain yield.
D Female black grouse mating with the male with the best display.

18. A population of finches became isolated on an island. The graph shows the range of beak sizes within the initial population at the time of isolation and in the population after many generations.



Which row in the table shows the type of selection pressure and the type of speciation which might be expected to occur in this example?

	Selection pressure	*Speciation*
A	directional	allopatric
B	directional	sympatric
C	stabilising	allopatric
D	stabilising	sympatric

[Turn over

19. Some processes involved in evolution are shown.

1 sexual selection

2 disruptive selection

3 genetic drift

Which of these processes involve non-random changes in the frequency of DNA sequences?

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

20. The analysis of DNA sequences from different organisms is used in the production of molecular clocks.

This analysis is based on the assumption that over time DNA sequences undergo mutations

A randomly

B spontaneously

C at a varying rate

D at a constant rate.

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

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National Qualifications 2018

Mark

X707/76/01

Biology
Section 1 — Answer Grid and Section 2

TUESDAY, 15 MAY
9:00 AM – 11:30 AM

X7077601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth
Day Month Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks
Attempt ALL questions.
Instructions for the completion of Section 1 are given on *page 02*.

SECTION 2 — 80 marks
Attempt ALL questions.
Questions 11 and 14 contain a choice.
Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.
Use **blue** or **black** ink.
Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X707760101

SECTION 1 — 20 marks

The questions for Section 1 are contained in the question paper X707/76/02.

Read these and record your answers on the answer grid on *page 03* opposite.

Use **blue** or **black** ink. Do NOT use gel pens or pencil.

1. The answer to each question is **either** A, B, C or D. Decide what your answer is, then fill in the appropriate bubble (see sample question below).

2. There is **only one correct** answer to each question.

3. Any rough working should be done on the additional space for answers and rough work at the end of this booklet.

Sample question

The thigh bone is called the

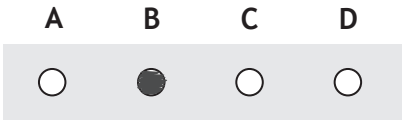
A humerus

B femur

C tibia

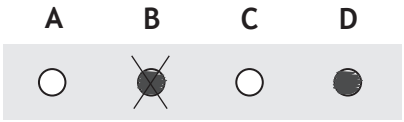
D fibula.

The correct answer is **B** — femur. The answer **B** bubble has been clearly filled in (see below).

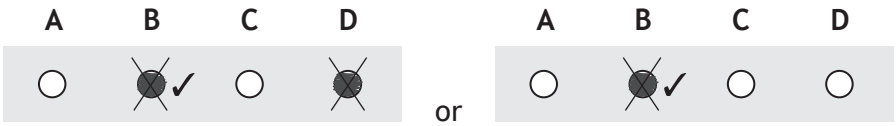


Changing an answer

If you decide to change your answer, cancel your first answer by putting a cross through it (see below) and fill in the answer you want. The answer below has been changed to **D**.



If you then decide to change back to an answer you have already scored out, put a tick (✓) to the **right** of the answer you want, as shown below:



SECTION 1 — Answer Grid

OBJ20AD1

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○
11	○	○	○	○
12	○	○	○	○
13	○	○	○	○
14	○	○	○	○
15	○	○	○	○
16	○	○	○	○
17	○	○	○	○
18	○	○	○	○
19	○	○	○	○
20	○	○	○	○

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[Turn over for next question

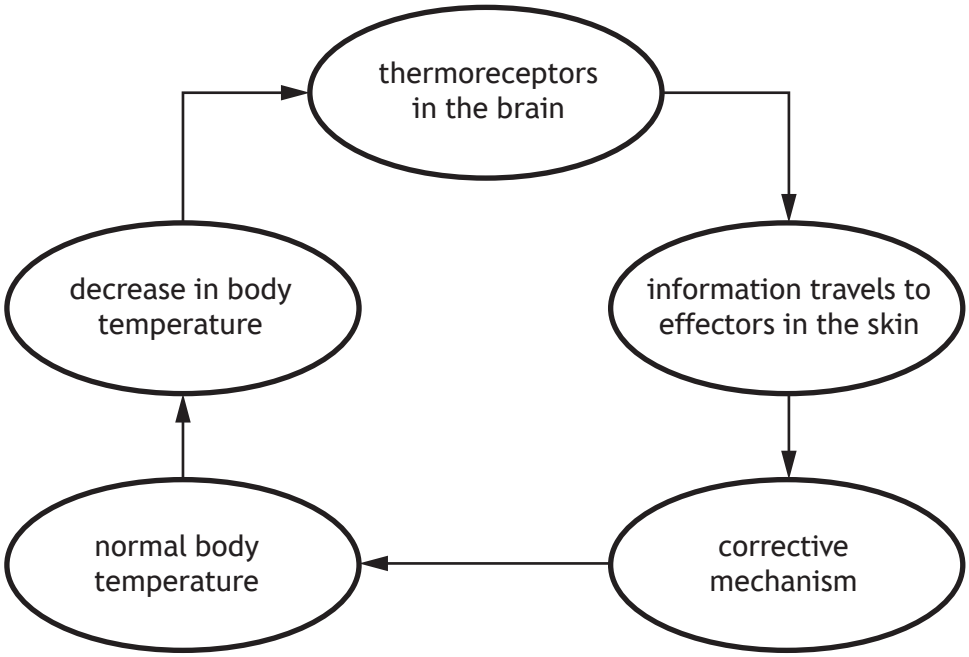
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SECTION 2 — 80 marks

Attempt ALL questions

Questions 11 and 14 contain a choice

1. The diagram illustrates thermoregulation in mammals following a decrease in body temperature.



(a) (i) Name the type of control used in thermoregulation as shown in the diagram. **1**

_____

(ii) Name the part of the brain in which thermoreceptors are found. **1**

_____

(iii) State how information travels to the effectors in the skin. **1**

_____

1. (continued)

(b) Effectors in the skin include muscles in the walls of blood vessels.

(i) Describe the response of these effectors to a decrease in body temperature. **1**

(ii) Explain how this response would help return body temperature to normal. **1**

(c) Explain why it is important for a mammal to regulate its body temperature. **1**

[Turn over

2. Daphnia (*Daphnia pulex*) is a species of water flea that lives in fresh water. An investigation was carried out into the effect of water temperature on the heart rate of one Daphnia. The results are shown in the table.

Water temperature (°C)	*Heart rate* (beats per minute)
2	175
7	184
12	194
17	207
22	219

(a) Calculate the average increase in heart rate per °C between 2 °C and 22 °C. **1**

Space for calculation

_____ beats per minute per °C

(b) Daphnia is a conformer. Use evidence from the table to confirm this statement. **1**

_____

_____

(c) Explain how an increased water temperature would result in a higher metabolic rate in Daphnia. **2**

_____

_____

_____

(d) State the type of response shown by conformers to maintain an optimum metabolic rate. **1**

_____

[Turn over for next question

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3. The bacteria *Streptomyces* is a microorganism found in soil. It produces a secondary metabolite, the antibiotic streptomycin, which kills other microorganisms. *Streptomyces* live in close association with plant roots. These plants produce soluble carbohydrates which are released into the soil through their roots.

(a) (i) Name the growth phase during which streptomycin is produced. **1**

_____

(ii) Explain the advantage to *Streptomyces* of producing an antibiotic such as streptomycin. **1**

_____

_____

(b) The relationship between *Streptomyces* and the plant roots is described as mutualistic.

(i) Suggest the benefit to *Streptomyces*. **1**

_____

_____

_____

(ii) Suggest the benefit to the plant. **1**

_____

_____

_____

3. (continued)

(c) An investigation was set up to compare the effectiveness of streptomycin with other antibiotics by measuring the survival of bacteria. A species of bacteria was grown in the presence of different concentrations of antibiotics and the percentage which survived was calculated. The results are shown in the table.

Antibiotic	*Concentration of antibiotic* ($\mu g/cm^3$)	*Survival of bacteria* (%)
Fusidic acid	10	6
Chloramphenicol	25	42
Erythromycin	5	49
Gentamycin	10	5
Tetracycline	25	35
Streptomycin	10	35

(i) Name one antibiotic with which streptomycin could be validly compared. **1**

_____

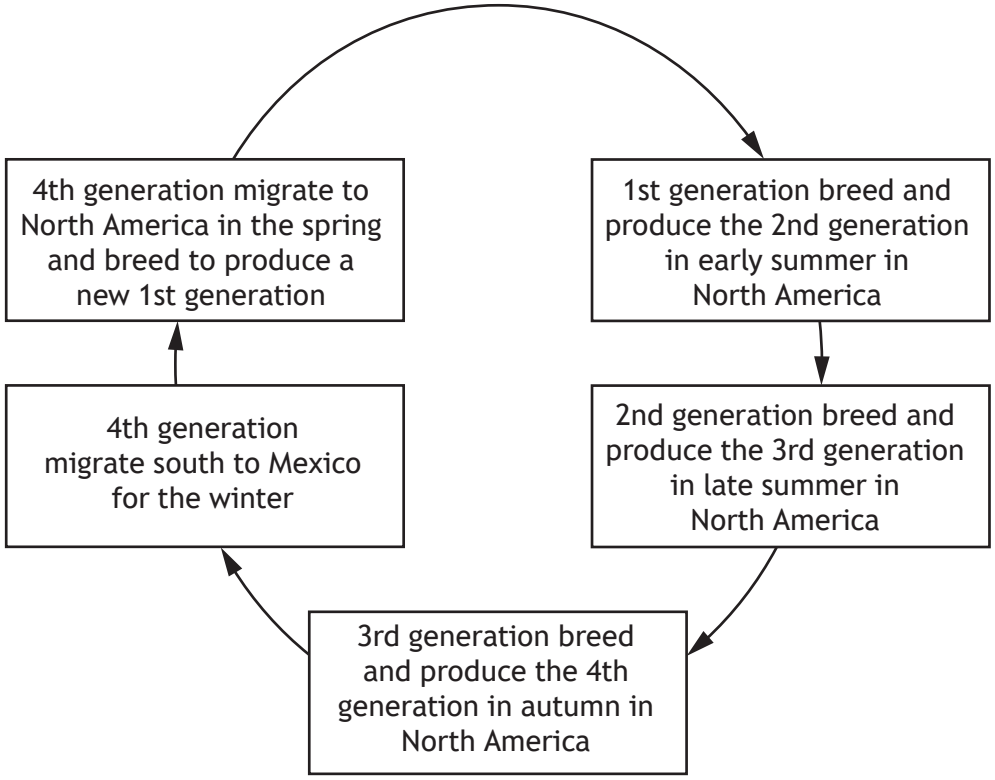
(ii) Give a conclusion which can be drawn from the results. **1**

_____

_____

[Turn over

4. The diagram shows information on the breeding and migration of Monarch butterflies (*Danaus plexippus*). Each generation dies after laying eggs.



(a) State one advantage and one disadvantage to the Monarch butterfly of migration to Mexico. **2**

Advantage _____

_____

Disadvantage _____

_____

4. (continued)

(b) The migratory behaviour of the Monarch butterfly from North America to Mexico is innate.

Use the information given to justify this statement. **1**

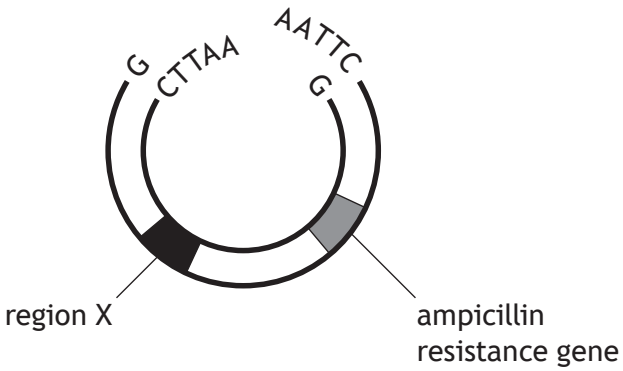
(c) Some species of hummingbird also migrate between North America and Mexico. They have high metabolic rates which they reduce while resting each night during the migration period.

Name this reduction in metabolic rate. **1**

[Turn over

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5. The diagram shows some features of a plasmid which has been cut open by a restriction endonuclease to allow a gene from a donor chromosome to be inserted.



The DNA recognition sites for three restriction endonucleases, *E coR1*, *BamH1*, and *HindIII*, are shown in the table. The arrows indicate where each restriction endonuclease cuts the DNA sequence.

Restriction endonuclease	*DNA sequence recognised*
E coR1	↓ G A A T T C C T T A A G ↑
BamH1	↓ G G A T C C C C T A G G ↑
HindIII	↓ A A G C T T T T C G A A ↑

5. (continued)

(a) (i) A restriction endonuclease was used to remove a gene from a donor chromosome.

Use information from the diagram and the table to identify the restriction endonuclease which would be used to allow the gene to be inserted into the plasmid.

Give a reason for your answer. **2**

Restriction endonuclease _____

Reason _____

_____

(ii) Name the enzyme which would be used to seal the gene into the plasmid. **1**

_____

(iii) A culture of bacterial cells, 20% of which had taken up this modified plasmid, were grown on a nutrient agar plate. The plate was incubated and 250 colonies of this bacteria grew.

Predict the number of colonies which would have been expected to grow if the nutrient agar plate had contained the antibiotic ampicillin. **1**

Space for calculation

_____

(b) Name region X, shown in the diagram, which ensured that the modified plasmid would be passed on to daughter cells. **1**

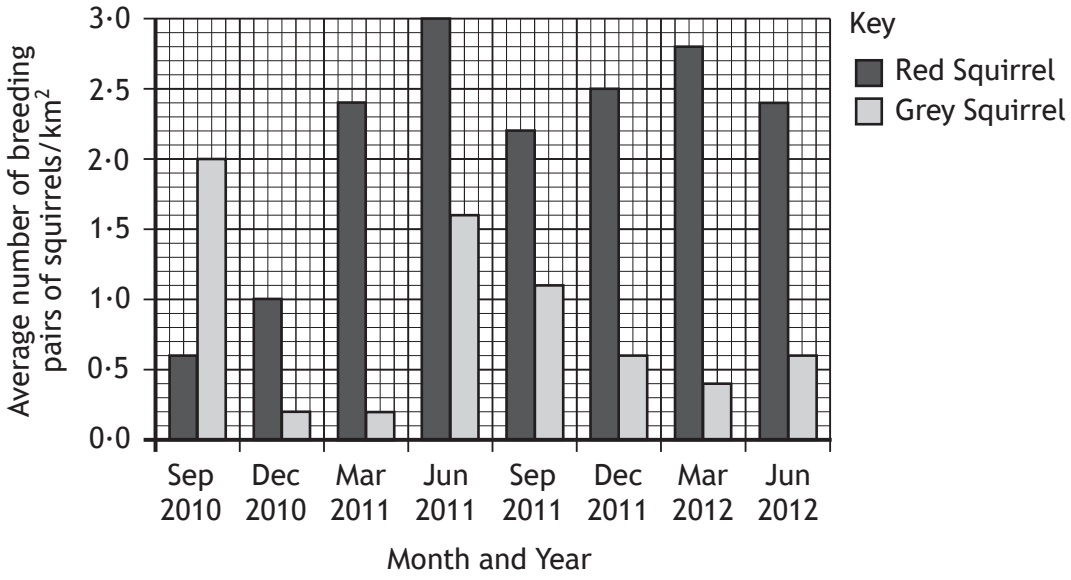
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[Turn over

6. An investigation was carried out to monitor the populations of red squirrels (*Sciurus vulgaris*) and grey squirrels (*Sciurus carolinensis*) in a 15 km^2 wooded area.

The average number of breeding pairs of each species was recorded between September 2010 and June 2012.

The results are shown in the graph.



(a) (i) Use **values from the graph** to describe the changes in the average number of breeding pairs of grey squirrels from March 2011 to March 2012. **2**

(ii) Calculate the **total** number of breeding pairs of red squirrels in the wooded area in September 2011. **1**

Space for calculation

_____ breeding pairs

(iii) Express, as the simplest whole number ratio, the number of grey squirrels to red squirrels in June 2012. **1**

Space for calculation

_____ : _____
grey red

6. (continued)

(b) Pine martens (*Martes martes*) live in wooded areas and prey on squirrels.

Populations of pine martens, red squirrels and grey squirrels were estimated in two other wooded areas using automatic cameras. These cameras are triggered by the movement of passing animals.

The results are shown in the table.

Wooded area	*Estimated number of grey squirrels*	*Estimated number of red squirrels*	*Estimated number of pine martens*
1	88	645	45
2	465	112	12

(i) Suggest why the method used to estimate the numbers of pine martens and squirrels may lead to inaccurate results. **1**

(ii) Use evidence from the table which could be used to support the following statements.

1 Pine martens are more successful predators of grey squirrels than of red squirrels. **1**

2 Grey squirrels compete more successfully for food than red squirrels. **1**

(c) Grey squirrels have spread rapidly and eliminated native red squirrels from much of the UK.

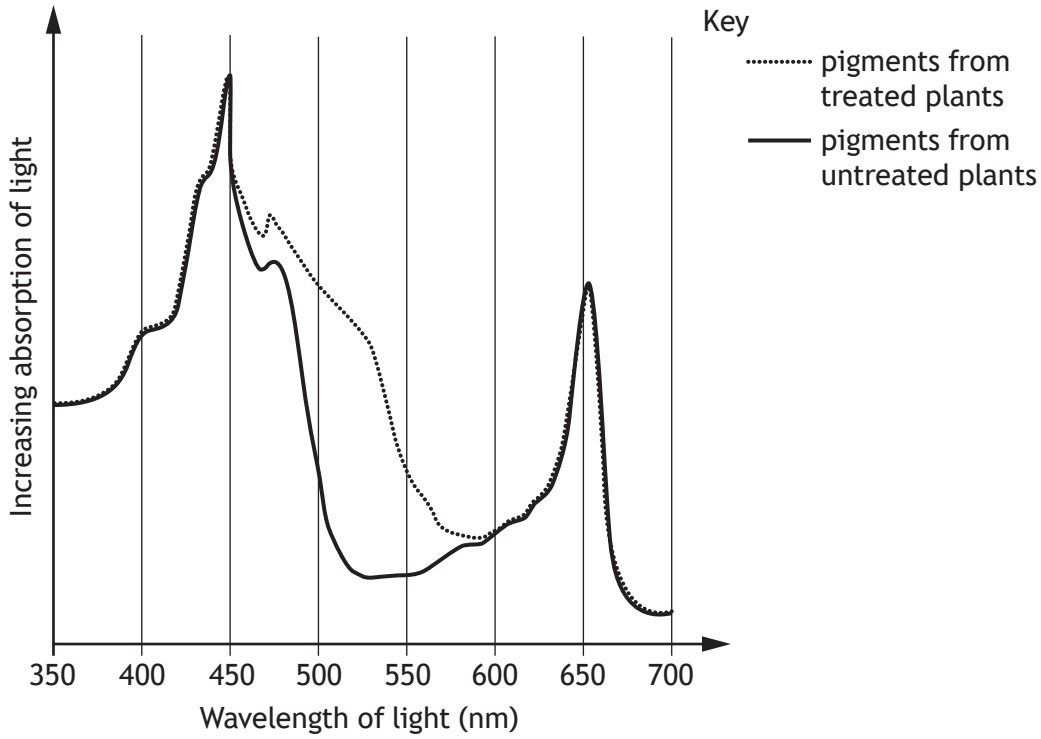
State the term used to describe grey squirrels in the UK as a result of this. **1**

[Turn over

7. Oil extracted from the seeds of the crop false flax (*Camelina sativa*) can be used as fuel. An investigation was carried out into the effect of a plant growth regulator paclobutrazol (PBZ) on the photosynthetic pigment content of the leaves and the oil yield from the seeds of false flax. The results are shown in the table.

Treatment	*Average photosynthetic pigment content* (mg/g of leaf)		*Average oil yield* (g/plant)
	Chlorophyll a and b	*Carotenoids*	
Untreated	3·28	3·02	1·7
Treated with PBZ	3·27	3·98	2·4

Absorption spectra for pigments from the treated and untreated plants were produced and are shown in the graph.



(a) (i) **Use values from the graph** to describe the difference in absorption spectra of the pigments from treated and untreated plants. **1**

_____

_____

_____

7. (a) (continued)

(ii) **Use evidence from the table** to explain the difference in the absorption spectra. **1**

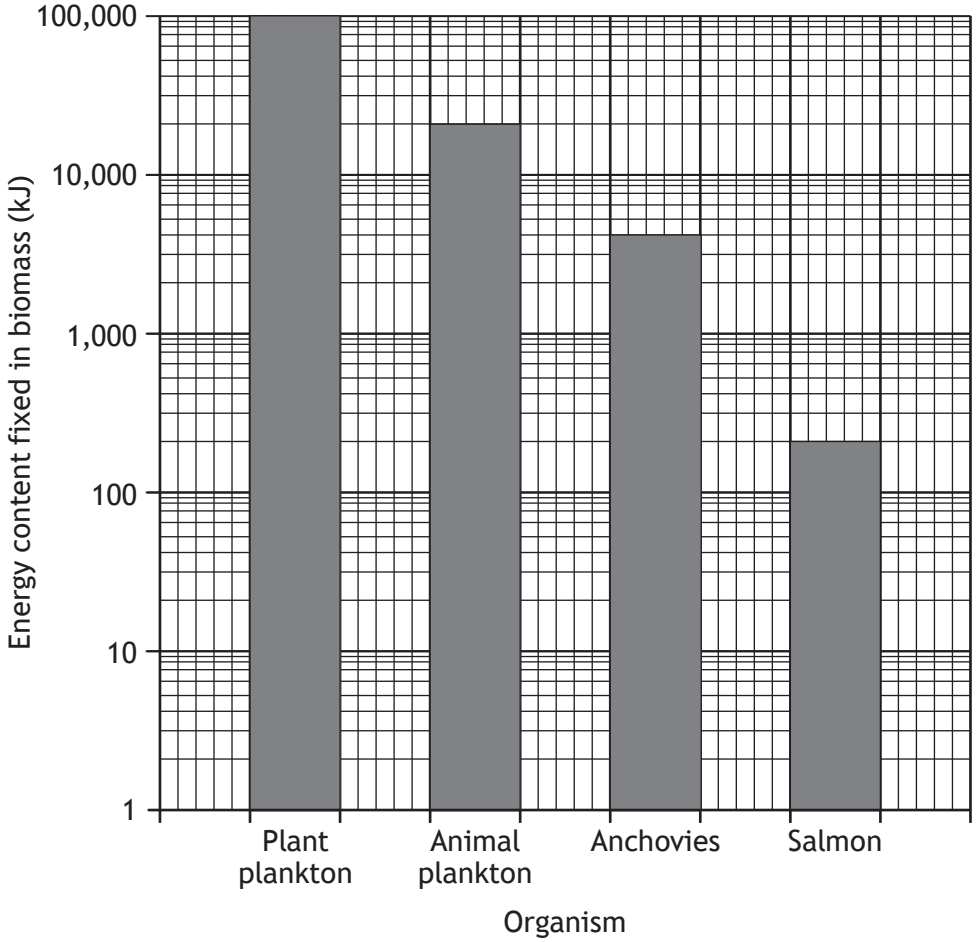
(iii) State one use that plants make of the light energy absorbed by pigments during photosynthesis. **1**

(b) Suggest why seeds from the plants treated with PBZ yield more oil. **2**

[Turn over

8. Salmon can be reared in fish farms where they are sometimes fed small fish such as anchovies. Anchovies feed on animal plankton which feed on plant plankton.

The energy contents at each trophic level in this food chain are shown on a log scale in the bar graph.



(a) (i) State the energy content fixed in the biomass of salmon. **1**

_____ kJ

(ii) Plant plankton fix 2% of the solar energy they receive in their biomass.

Calculate the **total** solar energy to which the plant plankton were exposed. **1**

Space for calculation

_____ kJ

8. (continued)

(b) The human population is increasing.

(i) State the term used to define the ability of the human population to access food of sufficient quantity. **1**

_____

(ii) In terms of energy explain the advantage to the human population of consuming anchovies rather than salmon. **1**

_____

_____

_____

[Turn over

9. The diagram shows crosses in a breeding programme involving different breeds of sheep.

(P) Scottish Blackface female X Border Leicester male

↓

Greyface female (F_1) X Suffolk male

↓

(F_2) Commercial lambs

(a) Suggest a reason why breeding programmes such as this include crossbreeding. **1**

(b) Explain why Greyface sheep are produced by crossbreeding Scottish Blackfaces with Border Leicesters instead of breeding F_1 Greyface sheep together. **1**

(c) To produce commercial lambs which show a desired dominant characteristic, Suffolk males homozygous for that characteristic are used.

(i) Name the type of cross used to identify if the genotype of the desired characteristic in Suffolk males is homozygous. **1**

(ii) Explain the importance of selecting a Suffolk male homozygous for the desired dominant characteristic. **1**

9. (continued)

(d) Spider lamb syndrome is a hereditary condition in sheep caused by a recessive deleterious allele which results in limb deformities.

State why inbreeding could cause an increase in the number of lambs born with this condition. **1**

_____

_____

[Turn over

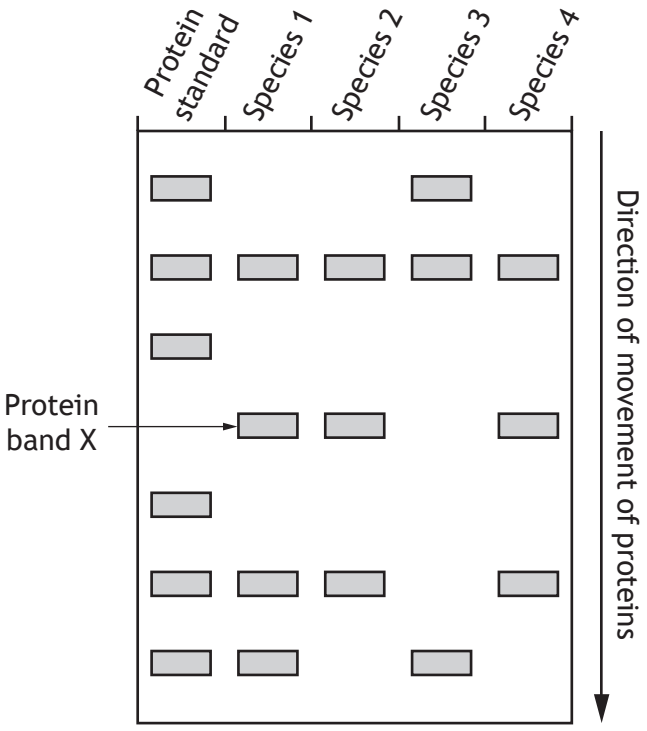
10. An experiment was carried out to investigate the evolutionary relatedness of four species of fish by comparing proteins extracted from the fish. The more closely related species are, the more proteins they have in common.

A sample of muscle tissue from each species of fish was heated in a solution to extract proteins.

The protein extracts were analysed by gel electrophoresis which separates proteins according to their mass. A protein standard containing proteins of known masses was also analysed.

The results of the gel electrophoresis are shown in the diagram. Each band represents a protein.

Mass of protein in protein standard (kDa)	*Distance travelled* (mm)
16	50
20	38
26	30
44	24
66	15
108	10



(a) (i) Identify **two** variables related to the protein extraction, not already mentioned, which should be kept constant so that a valid conclusion can be drawn. **2**

1 _____

2 _____

(ii) During the preparation the samples were heated. This unfolds the proteins changing their three-dimensional shape.

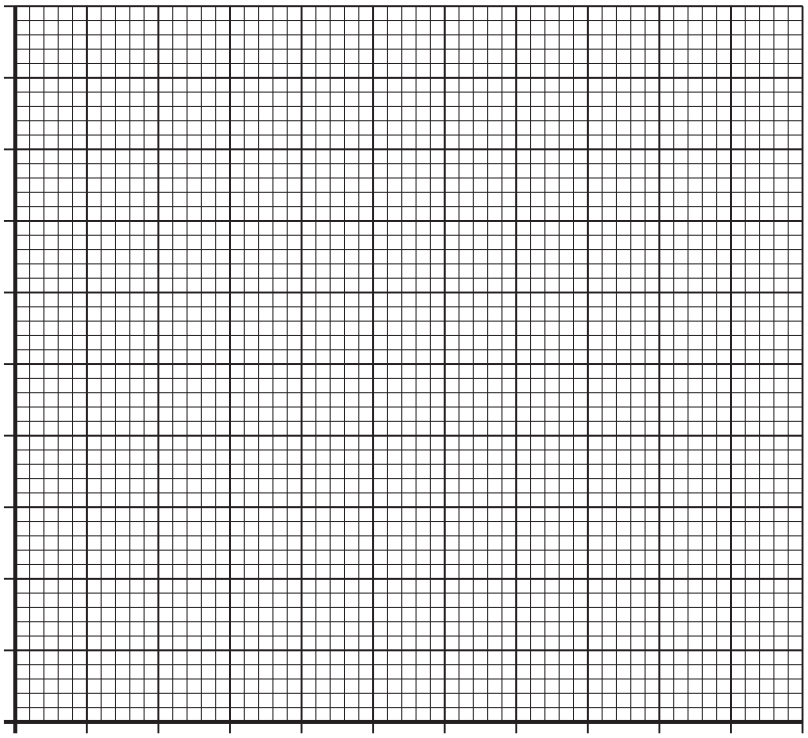
Name one type of bond that could have been broken to cause this change. **1**

_____

10. (continued)

(b) (i) Draw a line graph to show the distance travelled by the protein bands in the gel against the mass of protein in the protein standard. **2**

(Additional graph paper, if required, can be found on *page 33*)



Mass of protein in protein standard (kDa)

(ii) Band X travelled 28 mm. Use the graph to identify the mass of the protein in band X. **1**

_____ kDa

(iii) Each species of fish contains a protein with a mass of 66 kDa. One amino acid has an average mass of 0·12 kDa.

Calculate how many amino acids that would be expected in this protein. **1**

Space for calculation

_____

(iv) Explain why it was concluded that species 1, 2 and 4 are more closely related to each other than they are to species 3. **1**

_____

_____

11. Answer **either A** or **B**.

A Write notes on components of biodiversity and how these are measured. **4**

OR

B Write notes on parasitic relationships and transmission of parasites. **4**

12. Congenital lactase deficiency in humans is caused by very low activity of the enzyme lactase, resulting in individuals being unable to digest lactose in milk. This is caused by a number of different mutations in the lactase gene.

(a) One of the mutations involved causes a frame-shift mutation in the lactase gene.

(i) Name a gene mutation which causes a frame-shift. **1**

_____

(ii) Describe the effect of this frame-shift mutation on the lactase gene and on the structure of lactase. **2**

Effect on lactase gene _____

_____

_____

Effect on structure of lactase _____

_____

_____

(b) (i) Some mutations occur in the sequences which regulate the transcription of the lactase gene.

Suggest why this may lead to more lactase enzyme being produced. **1**

_____

_____

_____

(ii) Mutations in this gene are more common in Finland than in other parts of Europe. This is thought to be due to a small number of individuals who settled in Finland many generations ago.

State the term which describes this change in gene frequency when a small population of individuals breaks away from a larger population. **1**

_____

[Turn over

13. Scientists have used chemicals to produce polyploids to try to improve the characteristics of kiwi fruit plants.

The table shows some characteristics of fruit from the original diploid plant (2n) and two polyploids produced from it (4n and 6n).

	Characteristics of fruit			
Ploidy	*Average fruit mass* (g)	*Average fruit length* (mm)	*Average fruit diameter* (mm)	*Vitamin C content* (units)
2n	96·5	74·4	51·7	123·0
4n	115·8	76·1	60·0	119·0
6n	145·0	86·8	60·0	124·0

(a) (i) State which characteristic is least affected by polyploidy. **1**

_____

(ii) Describe the relationship between ploidy and average fruit diameter. **2**

_____

_____

_____

(b) Calculate the percentage increase in average fruit mass when the chromosome number is doubled. **1**

Space for calculation

_____ %

13. (continued)

(c) Describe the event that has occurred in the cells of the kiwi fruit plant that resulted in polyploidy. **1**

(d) Polyploids have whole genome duplications.

Explain the importance of this in evolution. **1**

[Turn over

14. Answer **either A** or **B** in the space below and on *pages 31* and *32*.

A Write notes on DNA under the following headings.

(i) Organisation of DNA in prokaryotic and eukaryotic cells; **4**

(ii) The polymerase chain reaction (PCR). **5**

OR

B Write notes on RNA under the following headings.

(i) Structure and functions of different types of RNA; **6**

(ii) RNA splicing. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

SPACE FOR ANSWERS

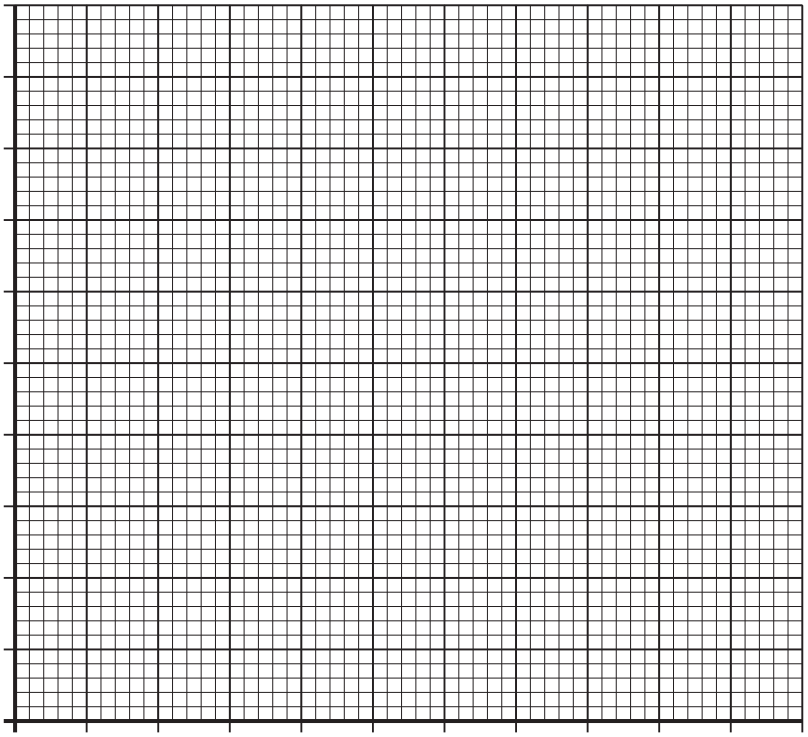
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[END OF QUESTION PAPER]

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional graph paper for question 10 (b) (i)



Mass of protein in protein standard (kDa)

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